

SHILKA, Russian Federation

WMO: 308620

Lat: 51.867N

Lon: 116.033E

Elev: 597

StdP: 94.36

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -35.8	(c) -33.5	(d) -38.3	(e) 0.1	(f) -35.3	(g) -36.2	(h) 0.1	(i) -33.1	(j) 9.2	(k) -16.1	(l) 8.2	(m) -17.2	(n) 1.5	(o) 270	(p) 0.668

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	31.0	18.1	28.9	17.4	27.0	16.9	20.0	26.9	19.1	25.3	18.2	24.2	4.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.9	13.8	22.2	17.1	13.1	21.7	16.1	12.3	20.8	60.5	27.2	57.2	25.4	54.3	24.1	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.7	9.2	7.9	-39.2	34.5	2.7	2.2	-41.1	36.1	-42.7	37.4	-44.2	38.6	-46.1	40.2
			-38.9	21.5	2.9	1.4	-41.0	22.5	-42.8	23.4	-44.4	24.2	-46.5	25.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.3	-23.8	-18.7	-8.6	2.8	10.7	18.0	20.3	17.6	10.6	0.8	-12.6	-22.1		
	DBStd	16.44	5.55	5.80	6.90	4.94	4.34	3.84	2.89	3.37	4.31	5.15	7.04	6.28		
	HDD10.0	4696	1047	802	578	222	43	1	0	1	44	286	677	996		
	HDD18.3	6954	1306	1036	836	466	239	50	13	53	234	542	927	1254		
	CDD10.0	935	0	0	0	6	66	242	320	237	62	2	0	0		
	CDD18.3	150	0	0	0	0	3	41	74	31	1	0	0	0		
	CDH23.3	1616	0	0	0	3	60	503	704	314	31	0	0	0		
	CDH26.7	503	0	0	0	1	12	168	229	89	4	0	0	0		
	WSAvg	3.3	2.7	2.9	3.5	4.6	4.5	3.6	3.1	3.0	3.3	3.4	2.9	2.5		
	PrecAvg	320	3	4	4	12	23	61	72	78	44	10	7	6		
(15) Precipitation	PrecMax	478	9	10	18	53	61	152	157	155	112	21	20	21		
	PrecMin	206	0	0	0	0	1	9	13	18	4	0	0	1		
	PrecStd	72	2	3	4	12	14	37	37	35	27	7	4	5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-7.9	-1.3	11.7	22.0	28.3	34.2	34.4	31.8	26.6	18.4	5.6	-4.0		
		MCWB	-9.5	-4.2	4.0	9.8	14.0	19.0	19.8	18.3	14.7	8.7	1.2	-6.0		
	2%	DB	-11.4	-5.0	7.4	17.7	24.7	30.6	31.4	29.2	23.4	14.2	2.5	-7.9		
		MCWB	-12.5	-7.0	1.4	7.2	12.4	17.2	18.7	17.8	13.3	6.5	-0.9	-9.2		
	5%	DB	-13.7	-7.8	4.1	14.3	22.1	28.3	29.0	26.8	20.7	11.3	0.0	-10.9		
		MCWB	-14.7	-9.4	-0.7	5.4	11.3	16.5	18.2	17.2	12.3	4.9	-2.5	-11.9		
	10%	DB	-16.1	-10.2	1.3	11.5	19.5	26.2	27.1	24.5	18.3	8.6	-2.7	-13.5		
		MCWB	-16.8	-11.4	-2.3	3.8	9.9	15.8	17.7	16.5	11.3	3.2	-4.7	-14.3		
	0.4%	WB	-9.3	-4.1	4.4	10.2	15.4	20.2	21.7	20.6	16.3	8.9	1.6	-6.0		
		MCDB	-7.9	-1.4	11.1	21.3	25.7	29.9	29.3	28.2	22.6	17.9	5.2	-4.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-12.5	-7.0	1.5	7.5	13.4	18.8	20.5	19.2	14.9	6.9	-0.6	-9.2		
		MCDB	-11.4	-5.0	6.9	16.5	22.3	26.7	27.5	25.1	20.0	13.5	2.2	-7.9		
	5%	WB	-14.6	-9.4	-0.5	5.7	12.0	17.8	19.6	18.4	13.6	5.1	-2.6	-11.9		
		MCDB	-13.7	-7.9	3.9	13.7	20.8	25.3	26.0	24.1	18.8	10.7	-0.1	-10.9		
	10%	WB	-16.8	-11.4	-2.3	4.0	10.5	16.8	18.8	17.5	12.2	3.6	-4.7	-14.3		
		MCDB	-16.1	-10.3	1.1	11.1	18.6	23.9	24.6	23.0	17.6	8.2	-2.8	-13.5		
	5% DB	MDBR	7.9	10.1	11.3	11.6	12.9	12.6	11.4	11.3	11.8	10.3	8.8	7.5		
		MCDBR	9.0	10.9	12.6	16.3	17.2	16.3	15.2	15.6	15.6	13.8	10.0	8.4		
		MCWBR	8.1	9.4	8.3	9.3	8.1	6.2	5.7	6.2	7.3	7.8	7.4	7.5		
		MCWBR	8.9	10.8	12.5	15.4	15.2	13.6	11.8	12.1	13.0	13.4	9.8	8.4		
(35) Mean Daily Temperature Range	5% WB	MCWBR	8.2	9.3	8.6	9.1	7.9	5.7	5.4	5.4	6.5	7.8	7.5	7.5		
	taub		0.249	0.276	0.317	0.416	0.493	0.467	0.419	0.388	0.344	0.306	0.275	0.254		
		taud	2.224	2.186	2.165	2.033	1.875	2.067	2.275	2.343	2.401	2.413	2.289	2.180		
(40) Clear-Sky Solar Irradiance	Ebn at Noon		767	842	870	813	763	792	829	838	839	806	723	679		
		Edh at Noon	66	93	117	152	187	156	125	111	93	74	62	58		
	RadAvg		1.46	2.63	4.24	5.23	5.81	6.13	5.61	4.88	3.85	2.60	1.50	1.11		
(44) All-Sky Solar Radiation	RadStd		0.04	0.06	0.18	0.31	0.38	0.52	0.41	0.30	0.30	0.16	0.07	0.04		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page